The Effects of Vouchers on School Results: Evidence from Chile's Targeted Voucher Program

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We use data from Chile's targeted voucher program to test the effects of vouchers on school results. Targeted vouchers have delivered extra resources to low-income, vulnerable students since 2008. Moreover, under this scheme, additional resources are contingent on the completion of specific education reforms. Using a difference-in-differences approach and a market-level empirical analysis, we find a positive and significant effect of vouchers on standardized test scores. Additionally, our results highlight the importance of conditioning the delivery of resources to some specific academic goals in markets with institutional characteristics that prevent public schools from behaving as profit-maximizing firms.

I. Introduction

A central question in education economics is how to increase the quality of schools that low-income students attend. Since Friedman wrote his 1955 essay on the role of government in education, many have considered the use of vouchers to be a promising way to increase the education quality supplied by schools. However, at the empirical level, the literature is far from reaching a consensus. This paper contributes to the literature by providing additional empirical evidence on the effects of vouchers on

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school results. We use data from a targeted voucher program introduced in Chile in 2008. A special characteristic of that program is that resources are conditional on compliance with certain scholastic goals by schools. We find a positive and significant effect of vouchers on school results. Additionally, our findings highlight the importance of conditioning the delivery of resources to the accomplishment of specific education reforms in markets in which institutional characteristics prevent public schools from behaving as profit-maximizing firms.

At a theoretical level, school vouchers can improve the academic results of low-income students through two main channels. First, vouchers allow low-income students to migrate from public to private schools. If private schools are better than public schools, vouchers should increase students' academic achievement just by allowing them to migrate from bad to good schools. Second, school choice can affect the overall system. Vouchers foster competition among schools, which can generate quality improvements in both public and private schools. However, empirical evidence from research on these two channels is not conclusive. Our paper focuses the analysis on the second channel.

Regarding the first channel, estimates of the achievement gains associated with private schooling often vary considerably even across studies that employ the same data sources. However, one result is robust across studies: Catholic schooling enhances educational attainment, especially for minority students in urban areas (Hill, Foster, and Gendler 1990; Evans and Schwab 1995; Figlio and Stone 1997; Neal 1998, 2002; among others).

Regarding the effects of school vouchers on the overall system, the results are also far from reaching a consensus. For instance, Rouse (1998), Brighouse (2000), and Hoxby (2001, 2003) argue that vouchers generate competitive forces that produce quality improvements in both public and private school systems. On the other hand, some studies, such as Hsieh and Urquiola (2006), find evidence of no improvements in average test scores of public schools after vouchers are introduced in the market.

In Chile, the empirical literature so far has exclusively used evidence on the universal voucher program implemented in 1981. These studies deliver mixed results that could be explained by differences in both data sources used and time period studied.

Regarding the first channel through which vouchers may increase the academic results of low-income students, early studies on private schooling use school-level data and find that, after properly controlling for students' socioeconomic backgrounds, public and voucher private schools perform similarly on achievement tests (Carnoy and McEwan 2000; Mizala and Romaguera 2000; Tokman 2002; Elacqua and Fabrega 2004). The availability of individual-level data since 1997 has induced a new generation of studies that include controls for students' resources and attempt to account for selection. Most studies using individual-level data find that students attending voucher private schools have higher educational outcomes than those from public schools (McEwan 2001; Mizala and

Romaguera 2001; Sapelli and Vial 2002, 2005; Lara, Mizala, and Repetto 2011; among others).

Regarding the effect of Chile's universal voucher program on the overall system, as in other countries, the evidence is not conclusive. For instance, using community-level data, Hsieh and Urquiola (2006) find that average standard test scores did not rise faster in communities that were exposed to more competition. However, other studies (Gallego 2006, 2012) find positive effects of competition on the overall system.[1]

In 2008 the Chilean government implemented a new voucher program targeted to low-income students that generated new data to study the effect of vouchers on school results. At first glance, this voucher program could be considered to be merely a more intense version of the 1981 universal voucher program. However, several differences exist between this reform and the one implemented in 1981.

The targeted voucher program is focused on low-income students, named *priority students*. Moreover, schools must sign the Equality of Opportunity and Educational Excellence Agreement to receive the extra resources delivered by the program. That agreement forces schools to present an Educational Improvement Plan to the Ministry of Education, which details education reforms that the school will undertake to improve results on the Sistema de Medición de la Calidad de la Enseñanza (SIMCE), a national standardized test.[2] Noncompliance with those reforms can result in the cancellation of the official recognition given by the Ministry of Education. Additionally, schools must detail how the extra funds delivered by the program fund will be spent to improve the academic performance of priority students and establish academic performance goals for their students, especially for those priority students. Schools that sign the agreement must also exempt all priority students from paying any copayment or tuition. Finally, schools must open their first- to sixth-grade admissions to any prospective student without taking into account past academic performance, current academic ability, or socioeconomic status. Therefore, schools that receive the extra funds from the program cannot select among priority students.

We use data from this voucher program implemented in Chile in 2008 to estimate the effect of vouchers on schools' results. We use two different empirical methodologies. First, we use a difference-in-differences approach. We build a panel of schools in which we can identify those that signed the Equality of Opportunity and Educational Excellence Agreement and received funds from the program from 2009 to 2011 (i.e., the

[1] In general, the scarcity of the literature on this second channel is due to the complexity of having both a voucher program covering a significant number of schools and the counterfactual markets.

[2] The SIMCE is a mandatory national standardized test designed to evaluate the quality of the content taught in primary and secondary education in math, language, geography, and science. It is administered annually to fourth, eighth, and tenth graders through a system in which grades are chosen to be evaluated by turns.

treatment group) and those that decided not to participate in this program during this period (i.e., the control group). The pretreatment period is defined as the time before the reform was implemented (i.e., from 2006 to 2008) and the posttreatment period as the time in which the reform was in effect (i.e., from 2009 to 2011). Second, we carry out a market-level analysis. We define each municipality as an individual market. Then we compare the changes in test scores after the implementation of the reform in markets in which a different number of schools signed the agreement.

Under both methodologies, we find a positive and statistically significant effect of vouchers on school results. After 3 years of exposure to the reform, the difference-in-differences approach shows that schools that participate in the program increase their math test scores by 0.4 standard deviations (with respect to the counterfactual school), which is equivalent to moving from the 50th percentile to the 65th percentile of the distribution. For language test scores, the total gain is 0.3 standard deviations, which moves a school from the 50th percentile to the 62nd percentile of the distribution. The market-level analysis shows that for each 1 percent of schools joining the program in a municipality, the average test score increases by 0.18 points in math and 0.11 points in language in that market. That means that in a market in which 80 percent of schools participate in the reform, the average gain is 14.4 points in math test scores and 8.8 points in language test scores after the third year, compared with a market in which no reform was implemented. Those effects are equivalent to test score gains of 0.53 and 0.37 standard deviations in math and language, respectively. The positive effect of the program on standardized test scores is robust to different alternative empirical specifications.

In environments in which public schools receive funds regardless of the number of students they attract, these schools may not react under increasing competition of an unconditional voucher system. Therefore, our results highlight the importance of conditioning the delivery of resources to some specific academic goals in markets with institutional characteristics that prevent public schools from behaving as profit-maximizing firms.

We discuss some potential channels through which the reform encouraged schools to improve quality. However, an important avenue for future research involves further empirical analysis to disentangle the mechanism through which the targeted voucher program changed the education market.

The rest of the paper is organized as follows. Section II reviews the existing literature on school vouchers. Section III describes the institutional details of the targeted voucher reform implemented in Chile. Section IV describes the data used. Section V discusses the empirical strategy used to test the effect of targeted vouchers on school results. Section VI presents the results. Section VII discusses our main results. Finally, Section VIII presents conclusions.

II. Related Literature

Research on school choice and vouchers typically focuses on two complementary research questions. First, how do school vouchers affect the outcomes of students who use the vouchers? Second, what is the effect of school choice on the overall system?

The first strand of the literature evaluates whether students who receive vouchers benefit from having the freedom to choose the school that they will attend. If private schools are better than public schools, vouchers should increase students' academic achievement just by allowing them to migrate from bad public schools to good private schools. Therefore, this strand of the literature is inevitably linked to the literature on the effects of private schooling.

The starting point was the 1981 report by James Coleman (Coleman, Kilgore, and Hoffer 1981), which concluded that Catholic and other private schools are, as a rule, more effective institutions of learning than public schools. Neal (1998) attempts to assess what we have learned since Coleman's report. He concludes that although many questions remain unanswered, one result seems clear. Black and Hispanic students in large cities often have the most to gain from private schooling, in particular, Catholic schooling. The poor quality of many inner-city schools appears to drive this result. In a subsequent work, Neal (2002) concludes that while it is difficult to predict the outcome of any large-scale voucher experiment, voucher systems targeted toward large cities with a history of public school failure may have the greatest potential for yielding large benefits.[3]

Another set of studies focuses on a more general comparison between public and both Catholic and non-Catholic private schools. In general, the evidence comes from small-scale voucher programs that are mostly designed for low-income students. Some programs are publicly funded, like the Milwaukee Parental Choice Program. Rouse (1998) uses data from the Milwaukee voucher program to measure the effect of vouchers on students who use the voucher. She provides evidence that access to private education increased the math scores of program participants, although she finds no evidence of positive effects on reading achievement. Using data from the same program in Milwaukee, Greene, Peterson, and Du (1998) perform a randomized experiment and confirm efficiency gains that can

[3] Additional studies supporting the idea that minorities in urban areas benefit from Catholic schooling are Evans and Schwab (1995), Hill et al. (1990), and Figlio and Stone (1997). Evans and Schwab (1995) use the National Center for Education Statistics' 1986 follow-up survey to their High School and Beyond study to examine the effects of private schooling. They find that Catholic schooling increases graduation rates. Additionally, they show that gains from Catholic schooling are larger for students in urban areas. Hill et al. (1990) support the claim that minority youths in large cities benefit from Catholic schooling. Figlio and Stone (1997) conducted an analysis using data from the 1988 National Educational Longitudinal Study. Correcting estimates for selection bias, they find that private schools with a religious affiliation do not enhance achievement in the population as a whole or within most subgroups. However, the authors do report large achievement gains for blacks and Hispanics who live in large, central cities.

result from privatization in education. Another publicly funded voucher program is the Washington, DC, Opportunity Scholarship Program. This program has been evaluated using a random assignment program design (Wolf et al. 2008). The evidence suggests at most small improvements in the academic results of students who move to private schools. Other voucher programs are privately funded, for instance, the New York City school voucher experiment. Using a randomized design, Mayer et al. (2002) and Krueger and Zhu (2004) report small but not statistically significant effects when all students are included and significant positive effects when only African American students are considered.[4]

Outside the United States, some studies have taken advantage of a randomized design. Angrist et al. (2002) present evidence on the impact of the Programa de Ampliacion de Cobertura de la Educacion Secundaria (PACES), a targeted and conditional voucher program in Colombia.[5] Their results suggest that PACES participants completed more years of school and had lower grade repetition, higher test scores, and a higher probability of working after graduating than students not enrolled in PACES. Angrist, Bettinger, and Kremer (2006) examine the longer-run effects of Colombia's PACES program. Their results show a substantial gain in both high school graduation rates and achievement as a result of the voucher program.

Angrist et al. (2006) discuss the issue of how to reconcile the consistently positive voucher effects that they found for Colombia with the more mixed results for the United States. They suggest three possible explanations. First, PACES could be a better experiment than those programs evaluated in the United States. Second, there is a larger gap in the quality of public and private schools in Colombia than in the United States. Finally, PACES included features not necessarily shared by other voucher programs, such as the incentives for academic advancement that introduce the conditionality of the program.

Although the studies of Angrist et al. (2002, 2006) suggest strong evidence of the overall effect of vouchers, they do not shed light on the underlying mechanisms through which the voucher effects emerged. Bettinger, Kremer, and Saavedra (2010) identified several plausible channels by which the Colombian voucher program has affected students' educational outcomes. These channels include income effects, changes in peers, changes in incentives, and changes in school resources. That study notes that the effects of school choice on those who exercise choice need not be

[4] A complete review of the literature on the impact of private-school vouchers can be found in Belfield and Levin (2002, 2003), Hoxby (2003), McEwan (2003), McEwan, Somers, and Willims (2004), Barrera-Osorio and Patrinos (2009), and Rouse and Barrow (2009). In the United States, a related literature studies the performance of charter schools. Hanushek et al. (2007) analyze the Texas charter experiment based on a panel of individual students who move across different schools, including charter schools. After controlling for student fixed effects to account for selection bias, Hanushek et al. find that after an initial start-up period, the performance of charter schools is statistically similar to that of public schools.

[5] Vouchers are renewed only for students who maintain satisfactory academic performance.

solely about the productivity of different schooling options. Understanding the channels through which vouchers enhance student achievement is important for formulating policy recommendations. Without knowing the mechanism, it is difficult to know whether the results will generalize to other voucher programs, as noted by Neal (2002).

In Chile, evaluations of the voucher program implemented in 1981 lack randomized designs. Therefore, studies mainly compare the achievement of students who attend public and private schools with controls for their observed and, more tentatively, unobserved characteristics. Achievement and socioeconomic data were available at the school level but not at the individual level until 1997; thus, early studies used the school as the unit of analysis.[6] Using school-level data, Carnoy and McEwan (2000) concluded that nonreligious private subsidized schools produce lower academic achievement than public schools, while Catholic private subsidized schools produce higher achievement outcomes than nonreligious private subsidized schools. Mizala and Romaguera (2000) argued that when sufficient control variables are taken into account, there are no consistent differences in achievement between public and private subsidized schools. Tokman (2002) finds that public schools are not consistently better or worse than private voucher schools, although public schools are more effective for students from disadvantaged family backgrounds. Elacqua and Fabrega (2004) find that students attending voucher private schools do not necessarily outperform public school students and that competition has not necessarily improved the test results of both types of schools.

Student-level analysis became possible when the Ministry of Education began to administer a questionnaire to all parents of students who participated in SIMCE, a standardized test in Chile. The availability of individual-level data since 1997 has generated new studies that control for students' resources and that attempt to account for selection. Most studies using individual-level data found that students attending voucher private schools have higher educational outcomes than those from public schools. McEwan (2001) finds that there is no consistent difference between student achievement in public and nonreligious private voucher schools, although fee-paying private schools and Catholic private voucher schools have higher achievement levels than public schools. Mizala and Romaguera (2001) and Sapelli and Vial (2002, 2005) find that students attending private voucher schools have higher educational outcomes than those from public schools. More recently, Anand, Mizala, and Repetto (2009) found that students in fee-charging private voucher schools score higher than students in public schools. However, they found no difference in the academic achievement of students in fee-charging private voucher schools relative to their counterparts in free private voucher schools. Their results suggest that low-income students who typically attend public schools can

[6] An empirical problem with this set of studies is that school-level data do not allow us to control for selection bias.

benefit from attending private voucher schools. Bravo, Mukhopadhyay, and Todd (2010) use the 2002 and 2004 Social Protection Surveys (Encuesta de Protección Social) to estimate a dynamic model of schooling and working decisions. They conclude that the school voucher program induced individuals affected by the program to attend private voucher schools at a higher rate, to achieve higher educational attainment, to participate more in the labor force, and to earn higher wages. Finally, Lara et al. (2011) use a number of propensity score–based econometric techniques and changes-in-changes estimation methods and find that private voucher education leads to small differences in academic performance.

Despite the favorable evidence presented in most of the Chilean studies that use individual-level data, it is difficult to determine the causes of the positive effects of private schooling. Some candidates could be better peers, superior teachers, more involved parents, and more effective management (in general, more productive schools). Overall, the evidence based on Chile's universal voucher program is less conclusive than the research on the Colombian voucher program, and conclusions depend heavily on the type of data used.[7]

In sum, the literature on the effects of private schooling delivers results that appear quite fragile. Estimates of the achievement gains associated with private schooling often vary considerably even across studies that employ the same data sources. However, there is one result that remains very solid across a number of studies: Catholic schooling enhances educational attainment, especially for minority students in urban areas.

The second line of research has attempted to identify the effect of school competition on the overall system. This strand of the literature focuses on the improvement of outcomes for all students in the system as a result of the increased opportunities for attendance. For instance, Rouse (1998), Brighouse (2000), and Hoxby (2001, 2003) argue that a voucher program fosters competition among schools, which generates quality improvements in both public and private school systems and promotes equality of educational opportunity.

In the United States, Hoxby (2003) analyzes the impact that different programs had on school productivity: specifically, the effect of vouchers on achievement in Milwaukee public schools and the effect of charter schools on achievement in Michigan and Arizona public schools. Hoxby concludes that the regular public schools boosted their productivity when

[7] There are three other potential reasons why research on the Chilean voucher program could have been less conclusive than the research on the Colombian voucher program. First, Chilean private schools can use selective admissions in accepting voucher students. As a result, Chilean voucher schools could have admitted students with better academic qualifications. Second, given that the program has existed for over 30 years and that a significant number of new private schools have entered the program, it is quite difficult to identify counterfactual outcomes of students who are attending schools that did not exist prior to the reform. Third, given the large magnitude of the program, the Chilean voucher program could have affected the overall quality of the system, and the control group in any regression specification may have also been influenced by the vouchers.

exposed to competition and that they responded to competitive threats that were surprisingly small.

In contrast to the programs analyzed in the United States, the universality of Chile's voucher program makes it especially suitable to analyze the effect of competition on school results. Hsieh and Urquiola (2006) use an identification strategy that compares communities that experienced a greater increase in private school enrollment to those that experienced less of an increase. Using community-level data, they find that average standard test scores did not rise faster in communities where the private-sector enrollment expanded more. Gallego (2006) uses the number of priests as an instrument for the penetration of the voucher program in a specific market. His findings suggest positive effects of the voucher program on the academic outcomes of students throughout municipalities where the voucher program had more penetration. His results may be indicative of competitive effects. More recently, Gallego (2012) studies the effects of interschool competition on the academic outcomes of Chilean students who attend publicly subsidized schools. He finds that, using instrumental variables, a one standard deviation increase in the ratio of voucher schools to public schools in a market increases tests scores by about 0.10 standard deviations.

Considering the differences in the time period studied is important to understand the differences in the results of the literature on Chile's universal voucher program. Gallego (2002) argues that the system operating in the 1980s differs in an important way from the system that has operated in recent decades. First, public school budgets were not affected by the voucher reform, in part because the decentralization of public schools was not completed until the late 1980s. Second, employment of public school agents was quite rigid until the mid-1990s. Third, test scores were not public until the late 1990s. Fourth, local governments were not elected democratically until 1992. Finally, the real value of the voucher decreased steadily during the 1980s and recovered only in the early 1990s.

There is also evidence of the competitive effects of vouchers in other countries of the world. The evidence from Sweden focuses on whether the program has improved outcomes throughout the entire education system or has exacerbated inequalities by increasing the amount of stratification between schools. As Bunar (2010) summarizes, even the positive estimates of the voucher program suggest that the voucher program may have had only a small impact on the overall education quality in Sweden. On the other hand, Bohlmark and Lindahl (2007) find that a 10 percentage point increase in the private school share increases average pupil achievement by almost 1 percentile rank point. Additionally, Bjorklund et al. (2004) and Bergstrom and Sandstrom (2005) both conclude that public schools improved as a result of competition from privately operated schools in Sweden.

For Canada, Card, Dooley, and Payne (2010) study the Ontario public education system to assess whether there is evidence of a willingness to switch schools and, if so, whether student performance is better in areas where the willingness to switch schools is greater. They test the prediction that test score gains between the third and sixth grades are larger for students in more competitive markets. The study finds that market characteristics associated with greater potential competition have statistically significant impacts on the growth rate of student achievement. The study's estimates suggest that expanding choice to all Ontario students would have a modest effect on sixth-grade test scores, raising achievement by 6–8 percent of a standard deviation.

In summary, in regard to the effects of vouchers on the overall system, the results are also far from conclusive. Although many studies have suggested positive effects in both Sweden and Chile, these studies employ identification schemes that are not perfect.

III. The Targeted Voucher Program

A. *The Chilean School System*

In 1980, Chile implemented several education reforms seeking to improve education quality and the efficient use of resources by fostering competition between schools. Before these reforms, Chile had a centralized education system whereby the vast majority of schools were publicly financed and the Ministry of Education was directly responsible for designing and overseeing the implementation of all education policies, both substantive and administrative. Additionally, there were subsidized private schools that were free for students and partially funded by the government but that received a lump-sum subsidy that was substantially smaller than per-student spending in the public sector.[8]

Three main changes were particularly transformative. First, administrative responsibility over public schools was transferred from the Ministry of Education to each municipal government, resulting in a more decentralized configuration. Second, a new scholastic subsidy system was introduced as the principal financing mechanism. Subsidized private schools began to receive exactly the same per-student payment as the public (municipal) schools. To distinguish these institutions from the subsidized private schools that existed before the reforms (mainly religious), we will call them voucher private schools. The subsidy was calculated as a function of total student enrollment and average student attendance. Third, the government moved to actively encourage the participation of private entities in the financing and administration of educational institutions. By making public funding directly contingent on student enrollment and retention, schools should compete for a larger share of the student body by offering a better education.

[8] At the beginning of 1980, more than 90 percent of all Chilean students were enrolled in institutions directly dependent on the Ministry of Education, and the remaining 10 percent were enrolled in completely private institutions.

In 1993, voucher private schools were allowed to charge students an additional tuition fee to complement the state subsidy.[9] However, some fraction of those voucher private schools continued not to charge any copayment to parents.

Therefore, since 1993, the Chilean primary and secondary education system has been composed of four types of schools: (i) public schools financed by government subsidies based on students' attendance and by additional funds from local governments (municipalities), (ii) free voucher private schools, (iii) fee-charging voucher private schools (which receive copayments from parents), and (iv) private schools financed exclusively by parents (private schools). Nonvoucher private schools are generally for-profit, whereas private subsidized schools can be either for-profit or non-profit. Nonvoucher private schools include both religious (mainly Catholic) and nonreligious schools.

There are 3.5 million students enrolled in the 12,500 schools that compose the Chilean school system. Of these, 5,670 are voucher private schools, 38.2 percent of which are fee-charging voucher private schools (which charge copayments to parents), whereas 61.8 percent are free private voucher schools. Additionally, 60 percent of fee-charging private voucher schools are for-profit schools. Overall, public schools, voucher private schools, and private schools represented 40.7 percent, 50.7 percent, and 8.6 percent, respectively, of the total enrollment in primary and secondary education in 2010.

Since the 1981 reform, more than 3,000 private voucher schools have been created. As a result, a massive migration from the public sector has occurred. Total enrollment in public schools decreased from 75 percent in 1982 to 40.7 percent in 2010. However, as shown by Hsieh and Urquiola (2006), despite extensive private entry and sustained declines in public enrollments, the aggregate number of municipal schools has barely fallen.

Currently, the kindergarten through grade 12 Chilean school system is divided into primary education (from kindergarten to eighth grade) and secondary education (from ninth to twelfth grades). Primary and secondary education are mandatory. Almost all private nonvoucher schools offer primary and secondary education. However, 88 percent of public schools and 61 percent of private voucher schools offer primary education only.

It is also important to note that there are important differences between public and voucher private schools. First, private and voucher private schools employed competitive admission processes to admit only the most promising applicants. In contrast, public schools with vacancies were legally obliged to admit all applicants regardless of their academic potential or socioeconomic backgrounds. Second, teachers' job contracts in public schools are regulated by a special legislation, the Teaching Statute, which involves a centralized collective bargaining process with wages based

[9] Subsidy law limited the family contribution to no more than US$153 per month (in 2012 values). The average copayment is US$30 per month.

on uniform pay scales and bonuses for training and experience. In contrast, private voucher schools hire and fire teachers according to the more flexible Labor Code. As a result, private voucher schools can more freely select, hire, and dismiss teachers, while municipal authorities find it a lot more difficult to fire teachers because of the Teaching Statute. Finally, public schools can receive municipal subsidies if the voucher is not enough to cover the entire budget. Because of the existence of those nonvoucher transfers, authors such as Sapelli and Vial (2002) claim that some public schools face a soft budget constraint.

B. The Targeted Voucher Program

Created in 2008, the targeted voucher program Subvención Escolar Preferencial (SEP) aims to improve the quality of education of the most vulnerable students by giving an additional per-student subsidy to schools that voluntarily enter the program. By providing additional resources to less advantaged students, named *priority students*, the additional subsidy aims to both improve the quality of education received by priority students and decrease socioeconomic inequality in the academic performance of students from different socioeconomic backgrounds, through a combination of individually allocated additional funds and an incentive-based school reform program. Moreover, for the first time, under this scheme, additional resources are contingent on the completion of specific scholastic reforms and improvements in school academic performance on a national standardized test (SIMCE).

1. Who Are Priority Students?

The fundamental basis of the program is the targeted support of a specific group of underprivileged and vulnerable students through additional subsidies. To automatically qualify as priority, a student must meet one of the following criteria in order, proceeding to further criteria if and only if the preceding one is not applicable: (i) participation in the Chile Solidario program (a social welfare program protecting those in extreme poverty), (ii) being in the most vulnerable third of the population according to the latest measurement instrument, or (iii) belonging to the most vulnerable group in the National Health Fund (FONASA). If none of the preceding criteria are met, the student's position can also be temporarily established according to his or her family socioeconomic background.

2. The Equality of Opportunity and Educational Excellence Agreement

To join the SEP program, schools must sign the Equality of Opportunity and Educational Excellence Agreement (Convenio de Igualdad de Oportunidades y Excelencia Educativa) with the Ministry of Education. This agreement is valid for 4 years and may be renewable for an equal period of time. During this period, schools agree to report the use of all resources received under the program and the status of specific projects. They must also present an Educational Improvement Plan to the Ministry of Education, which details education reforms that the school will undertake to improve academic results. Additionally, schools must detail how the extra funds delivered by the program will be spent to improve the academic performance of priority students and establish academic performance goals for their students, especially for priority students.

Schools that sign the agreement must exempt all priority students from paying any copayment or tuition that could condition their application, entry, or stay in the schools;[10] must open their first- to sixth-grade admissions to any prospective student without taking into account past academic performance, current academic ability, or socioeconomic status; and must retain all students, even those with poor academic performance.[11] These requirements for participation in the program are designed primarily to ensure that high-performing schools are accessible to all low-income students and that schools are not allowed to give preference to highly qualified priority students.

The Educational Improvement Plan contains specific actions and measures that schools will undertake in the following areas, emphasizing those that schools identify as needing the greatest improvements:

- Curriculum management: Schools should carry out measures to strengthen the school education project, improve pedagogic practices, support students with special needs, and improve student evaluation systems.
- School leadership: Schools should, for example, prepare and train school management teams, strengthen the value and civic formation of students, and involve the school in the community.
- Student life: Schools must take measures to provide psychological support and social assistance to students and their families and improve the relationships between students, their parents, and teachers.
- Resource management: Schools should take actions to define a policy to improve teacher performance; strengthen those areas of the curriculum in which students obtain unsuccessful results; design and implement evaluation systems for teachers; implement performance incentives for management teams, teachers, and administrative personnel (which must be linked to the goals established in the Edu-

[10] Despite the fact that schools that participate in the program are prohibited from charging any tuition or copayment to priority students, they can charge nonpriority students. This is especially relevant for voucher schools that do not lose the copayment of nonpriority students after they decide to participate in the program.

[11] If the demand surpasses the number of available spots, students will be admitted through a public and transparent application process that will not consider their socioeconomic status or their past or potential academic performance.

cational Improvement Plan); and foster instruments to support educational activities, such as a library, computers, and Internet access.

The actions included in the Educational Improvement Plan could be extraordinarily modified if the conditions that were used to define the plan changed over time. Additionally, the SEP law establishes that in order to accomplish the actions in the areas previously mentioned, schools can hire teachers, education assistants, and the necessary staff to improve their technical and pedagogical performance and to elaborate and develop the Educational Improvement Plan.

The Ministry of Education classifies schools into three categories according to their academic performance on a national standardized test (SIMCE) and the socioeconomic characteristics of their students. Depending on their performance on the SIMCE, schools could be classified as *autonomous* (if their performance is at or above the median for their SIMCE group), *emergent* (if their performance is below the median for their SIMCE group), or *in recuperation* (applied to emergent institutions that fail, after 4 years, to meet the quantitative goals required by the program). Although the amount of resources distributed to each priority student is the same for all schools and varies only by educational level, the autonomy of schools to decide how to use those resources and under what level of supervision will depend on the school's classification.[12]

The sanctions that schools could receive as a result of noncompliance with their Educational Improvement Plan depend on their classification. Autonomous and emergent schools could be demoted to a lower category (emergent and in recuperation, respectively) if they do not accomplish the initiatives included in their improvement plan. If in-recuperation schools fail to achieve the academic results of emergent schools or do not accomplish the measures established in their improvement plan, the National Education Quality Agency will offer to students' families the possibility of

[12] For instance, emergent schools receive half of their funding as a subsidy and the other half as a contribution of additional resources to design and execute their Educational Improvement Plan, and schools that are in recuperation receive all their funding as a contribution of additional resources to their plan. Additionally, emergent and in-recuperation schools must assume additional commitments. Emergent schools must include in their Educational Improvement Plan a diagnostic of the school's initial situation, in terms of its technical, human, and material resources, and a set of goals for educational results for the lifetime of the plan. Emergent schools must coordinate their actions with social institutions to detect and treat psychological and social problems and address the specific educational needs of priority students. Those schools must also establish complementary teaching and learning activities for priority students to improve their academic performance. On the other hand, the Educational Improvement Plan for in-recuperation schools must contain specific initiatives in administrative and management areas and those concerning students' learning process and must be elaborated before the beginning of the school year that follows the year when schools sign the Equality of Opportunity and Educational Excellence Agreement. To elaborate their improvement plan, in-recuperation schools could be assisted by the Ministry of Education and agencies certified by the Ministry of Education.

switching to a different school, and schools may lose the official recognition given by the Ministry of Education.[13]

3. Funds

SEP funding is allotted per priority student and is delivered directly to the school instead of to any municipal funding system. It is calculated on the basis of the average attendance rate of priority students over the previous 3 months, the grade level of the priority student (younger students receive more resources), and the concentration of priority students in the school. Table 1 shows the monthly subsidy delivered per priority student in 2012.

In addition to the regular funds received from the program, schools qualify for a subsidy if priority students make up more than 15 percent of the student body. Table 2 presents the concentration subsidy amount.

In relation to the flat voucher delivered by the 1981 reform, the program increases the resources given to schools for priority students by 70 percent. Table 3 presents the amount of additional resources given by the targeted voucher program.

IV. Data

Our sample is composed of public and voucher private schools that do and do not sign the Equality of Opportunity and Educational Excellence Agreement. The database contains information about students' socioeconomic backgrounds, including variables such as gross family income and the education level of the students' parents. We link this information with school academic performance. In particular, we include the yearly average score that a school obtained on the SIMCE during the period 2006–11. Considering that the reform was progressively implemented in preschool and primary education (and only since 2012 in secondary education), we use schools' average scores obtained by fourth graders in math and language, which are the only measures of school performance available for the levels under the program.

Using this information, we build a panel for public and voucher private schools that allows a pretreatment and a posttreatment period. We define the pretreatment period as the three years before the implementation of the SEP reform (2006–8). The posttreatment period corresponds to the three years in which schools could receive the extra funds by signing the Equality of Opportunity and Educational Excellence Agreement (2009–11). By signing the agreement, schools agreed to carry out specific measures designed to improve the quality of the education provided and perform specific actions to encourage the school's retention and academic performance of priority students.

[13] The National Education Quality Agency is in charge of evaluating and guiding the education system so that it leads to the improvement of the quality and equity of educational opportunities.

TABLE 1
Monthly Subsidy per Priority Student (US$)

	Prekindergarten to 4th Grade	5th and 6th Grades	7th–12th Grades
Monthly per-student subsidy	86.6	57.6	29.1

Source.—Ministry of Education, Chile.
Note.—Values are for 2012 and are adjusted by purchasing power parity.

TABLE 2
Monthly Concentration Subsidy per Priority Student (US$)

Concentration of Priority Students	Prekindergarten to 4th Grade	5th and 6th Grades	7th–12th Grades
15%–30%	6	4	2
30%–45%	10.3	6.9	3.4
45%–60%	13.8	9.2	4.6
≥60%	15.4	10.3	5.2

Source.—Ministry of Education, Chile.
Note.—Values are for 2012 and are adjusted by purchasing power parity.

TABLE 3
Flat Voucher and SEP for Prekindergarten and Fourth Graders (US$)

Category	Subsidy
Flat voucher	143
SEP	86.6
Concentration subsidy	13.8
Total	243.4

Source.—Ministry of Education, Chile.
Note.—Values are for 2012 and are adjusted by purchasing power parity.

From a total of 12,117 schools in 2009, we select all those schools with primary education that appear on SIMCE records from 2006 to 2011. From this sample, we select schools with 20 or more students taking the SIMCE, considering that the evaluation and test scores are not representative for schools with fewer than 20 students. Additionally, for treated schools, we select those that participated without interruption in the program from 2009 to 2011.

We use school-level data because data availability makes it impossible to perform an individual-level analysis. SIMCE evaluates students from the fourth, eighth, and tenth grades by turns. Although test scores are available at the student and school levels, we use schools as observational units because of the impossibility of following the same cohort of students during the pre- and posttreatment periods. However, since 2006, the SIMCE has evaluated fourth graders each year, which allows us to use school average SIMCE scores for schools that have and have not participated in the reform and to compare their academic performance using the methodology described in the next section. Additionally, if the same cohort of students were evaluated each year, it would not be possible to identify priority students and their transfers across schools since the available data contain only aggregate information on the percentage of priority students in a school.

V. Empirical Strategy

The Chilean targeted voucher program is not a randomized experiment, which makes the policy evaluation of the program difficult. In this section, we discuss two alternative empirical approaches to estimate the causal effect of the reform on the academic achievement of schools.

We first use a difference-in-differences approach. According to this methodology, there are two groups observed in two time periods, where one is exposed to a treatment in the second period but not in the first (treatment group). The other group is not exposed to a treatment during either period (i.e., the control group). If the same units of the treatment and control groups are observed in both the first and second periods, the effect of the treatment could be obtained by subtracting the average gain of the control group from the average gain of the treatment group. This approach allows us to remove the biases that result from the comparison of both groups in the second period and that can be explained by permanent differences between them and differences across time due to group-specific trends.

We use a panel of schools in which we can identify those that signed the Equality of Opportunity and Educational Excellence Agreement and received funds from the program from 2009 to 2011 (i.e., the treatment group) and those that decided not to participate in this program during this period (i.e., the control group).[14] In our model, the pretreatment period is defined as the time before the reform was implemented (i.e., from 2006 to 2008) and the posttreatment period as the time in which the reform was in effect (i.e., from 2009 to 2011). We estimate the following empirical model:

$$Y_{it} = \beta_1 \lambda_t + \beta_2 w_{it} + c_i + u_{it}, \tag{1}$$

where Y_{it} corresponds to the average score that school i obtained on the SIMCE (in math and language, depending on the econometric specification) in period t, λ_t is a time trend, and w_{it} is a binary program indicator that equals one if school i participates in the program at time t and equals zero otherwise. Finally, we include a fixed school effect c_i. The term u_{it} represents an idiosyncratic error term.

[14] It is important to note that schools enrolled in the program receive SEP funds for each enrolled priority student (to whom schools cannot charge copayments). However, those schools are free to enroll nonpriority students, for whom schools do not receive funds from the program and to whom they can charge copayments.

We estimate equation (1) by ordinary least squares, clustering by municipality to estimate standard errors. The coefficient β_2 of the previous regression estimates the average test score gains observed in schools that participate in the program.

An additional question is whether the program has a cumulative effect on school academic results, considering that, in our sample, all schools signed the agreement in 2009 and remained in the program until 2011. To explore whether the program had an additional effect for each additional year, we include an interaction of the policy indicator and a trend variable λ_t' that takes the value of zero during the pretreatment period and the values of one, two, and three during the posttreatment period for both treatment and control groups. By doing this, we capture the average effect of each additional year of exposure to the reform. Specifically, we run the following regression:

$$Y_{it} = \alpha_1 \lambda_t + \alpha_2 \lambda_t' w_{it} + c_i + u_{it}, \tag{2}$$

where the coefficient α_2 measures the average annual effect that the program had on participating schools.

Finally, to analyze the convexity or concavity of the test score gains, we implement an even more flexible empirical approach by including three different policy indicators for each year of the treatment period. Specifically, we estimate the following model:

$$Y_{it} = \delta_1 w_{i,2009} + \delta_2 w_{i,2010} + \delta_3 w_{i,2011} + \Delta D_t + c_i + u_{it}, \tag{3}$$

where D_t is a vector of time dummies, Δ is the vector of coefficients of those dummies, and the coefficient δ_j estimates the additional effect of the program after year j.

The key identifying assumption of a difference-in-differences approach is that the trend in test scores would be the same in both treatment and control schools in the absence of treatment. In that case, we interpret the policy effect estimator as the average effect of the reform on a school's academic results. Figure 1 shows test scores in the treatment and control groups. We observe that the trends followed by those school groups are fairly similar until 2008.

If the education market is characterized by competition for shares, the parallelism in the absence of the program implied by the difference-in-differences assumption is hard to support. When control schools "react" to changes in the quality offered by treatment schools, it is difficult to argue that the test score trend followed by the control group represents a good counterfactual for the trend that treatment schools would have followed in the absence of the treatment. In that case, the ideal methodology is to use data at the market level. That is, we can base identification on "closed" markets and then shock one or more schools with the program and look at market-level impacts on different types of children. We can identify specific locations where one or more schools joined the

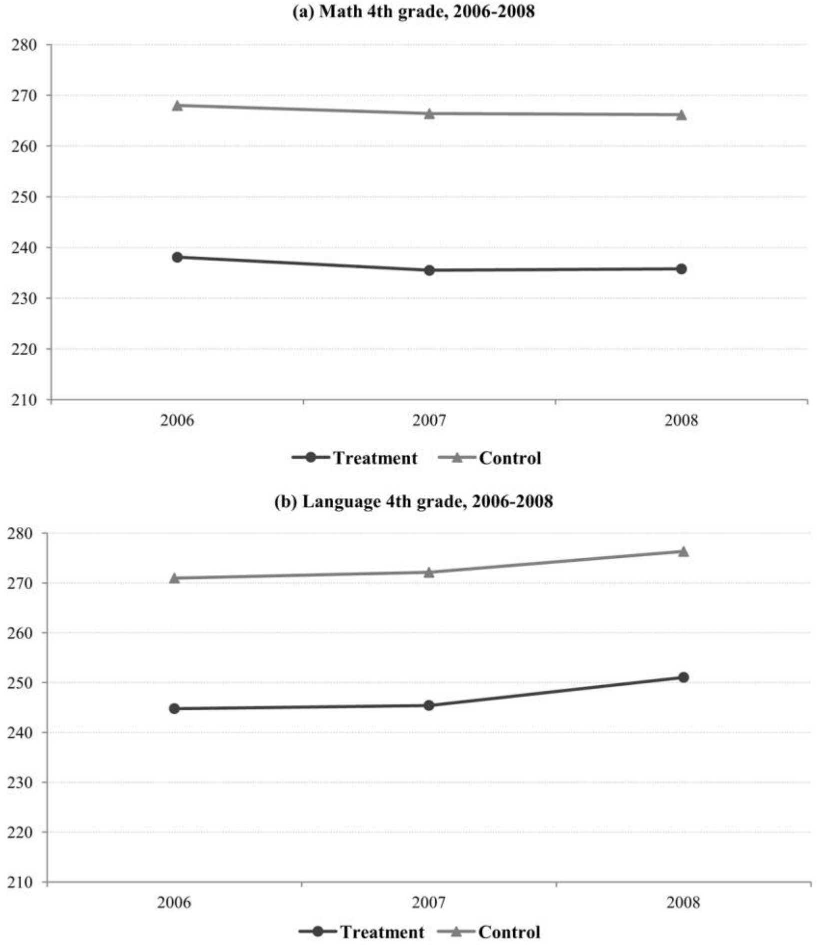

Figure 1.—Test scores in the treatment and control groups

program and compare them to other locations where schools were eligible but did not opt in. The outcome variable would then be the aggregate scores across all schools in the market. To follow such a strategy requires the identification of the control group, which is not straightforward.

We decided to pursue an alternative methodology. We define each municipality as an individual market. We also perform the analysis by defining some provinces as individual markets. Then we compare the changes in test scores after the implementation of the reform in markets in which a different number of schools signed the agreement. This methodology allows us to better capture the competitive forces that encourage both treatment and control groups to supply a higher level of quality. Specifically, we estimate the following empirical model:

$$\Delta Y_i = c + \alpha_1 S_i + \alpha_2 X_i + u_{it}, \tag{4}$$

where ΔY_i corresponds to the change in the average test score in market i since the treatment, S_i is the fraction of schools that join the program in market i, and X_i are covariates that include the change in the average test score and the average socioeconomic variables of market i during the pretreatment period. The term u_{it} represents an idiosyncratic error term, and c is a constant.

The next section presents the results of our two empirical approaches and discusses additional methodological issues.

VI. Results

Table 4 presents the results of equation (1). Columns 1–8 show a positive and statistically significant effect of the reform on math and language test scores. According to the basic specification 1, where no socioeconomic variables are included, schools that joined the program in 2008 experienced an average gain of 4.8 points in math test scores and 2.8 points in language test scores. Those effects are equivalent to test score gains of 0.18 and 0.12 standard deviations in math and language, respectively.[15] Despite the fact that the point estimate associated with math is higher than that for language, the trend coefficient associated with language is higher than that for math. This could be explained by reforms carried out to increase the quality of the language education content overall. As a result, test scores for language showed a tendency to increase, which can be attributed to curriculum reforms that were not implemented in math. Columns 2–4 and 6–8 show the estimated coefficients for math and language when family background is controlled for. The conclusions are similar to those reported in the baseline specification.

In table 5 we present the results from empirical model (2), that is, the specification that includes the interaction of the policy indicator and a trend variable λ_t' that takes the value of zero during the pretreatment period and the values of one, two, and three during the posttreatment period for both treatment and control groups. As discussed in Section V, with this empirical specification, we intend to capture the average effect of each additional year of exposure to the reform. Table 5 indicates that an additional year increases test scores by 4.1 points in math and 1.6 points in language, values that are statistically significant at the 1 percent level. Therefore, a school that has received funds from the program for 3 consecutive years improves its test scores by 12.3 points in math and 4.8 points in language, relative to their counterfactuals. These effects are equivalent to test score gains of 0.5 and 0.2 standard deviations in math and language, respectively. After including school socioeconomic background

[15] To obtain the equivalence of estimated effects in terms of standard deviations, we use the distribution of test scores for the whole sample of schools and calculate its standard deviation. Then, using simple proportion rules, we find the equivalence of the estimated effects of the reform on test scores in terms of standard deviations for the different regressions presented in this section.

TABLE 4

EFFECTS OF THE SEP REFORM ON SCHOOL ACADEMIC ACHIEVEMENT

	SIMCE 4th-Grade Math				SIMCE 4th-Grade Language			
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
λ_t	.8632***	.7382***	.8274***	.9130***	2.4463***	2.2876***	2.4144***	2.4160***
	(.133)	(.145)	(.127)	(.143)	(.111)	(.111)	(.107)	(.108)
w_{it}	4.7634***	4.9283***	4.4684***	4.3418***	2.8204***	3.0297***	2.5897***	2.5872***
	(.432)	(.433)	(.419)	(.421)	(.433)	(.430)	(.429)	(.426)
Gross family income		.0100***		−.0069**		.0128***		−.0001
		(.003)		(.003)		(.003)		(.003)
Mother's years of schooling			3.2819***	3.3743***			2.7572***	2.7589***
			(.289)	(.295)			(.259)	(.263)
Father's years of schooling			2.0068***	2.1285***			1.4507***	1.4530***
			(.256)	(.261)			(.249)	(.253)
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Source.—SIMCE 2006–11 and Ministry of Education, Chile.

Note.—N = 18,203. Gross family income corresponds to the average family income that students report at the moment of taking SIMCE. It is expressed in thousands of Chilean pesos (CLP) for each one of the years considered in the sample (1 US$ = 520 CLP). Standard errors (in parentheses) were estimated using clustering by municipality

* $p < .1$.

** $p < .05$.

*** $p < .01$.

TABLE 5

Cumulative Effects of the SEP Reform on School Academic Achievement

	SIMCE 4th-Grade Math				SIMCE 4th-Grade Language			
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
λ_t	−.1639	−.4129***	−.1362	−.1437	2.2270***	2.0051***	2.2470***	2.2200***
	(.117)	(.124)	(.114)	(.126)	(.113)	(.114)	(.111)	(.113)
$\lambda_t' \times w_{it}$	4.0698***	4.2698***	3.8264***	3.8331***	1.6065***	1.7848***	1.4076***	1.4317***
	(.216)	(.211)	(.212)	(.208)	(.206)	(.205)	(.208)	(.208)
Gross family income		.0166***		.0005		.0148***		.0018
		(.003)		(.003)		(.003)		(.003)
Mother's years of schooling			3.1524***	3.1457***			2.7151***	2.6908***
			(.291)	(.298)			(.259)	(.263)
Father's years of schooling			1.8815***	1.8727***			1.4235***	1.3918***
			(.254)	(.258)			(.248)	(.252)
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Source.—SIMCE 2006–11 and Ministry of Education, Chile.

Note.—$N = 18{,}203$. Gross family income corresponds to the average family income that students report at the moment of taking SIMCE. It is expressed in thousands of Chilean pesos (CLP) for each one of the years considered in the sample (1 US$ = 520 CLP). Standard errors (in parentheses) were estimated using clustering by municipality.

* $p < .1$.

** $p < .05$.

*** $p < .01$.

variables (family income and parents' schooling), results remain positive and significant.

To gain a better idea of the impact of the reform, we can consider a school in the 50th percentile of the test score distribution. An increase of 12.3 points would place the school in the 66th percentile in the math score distribution, while an increase of 4.8 points would locate the school in the 58th percentile in the language score distribution.

Additionally, we also estimate the more flexible empirical model (3) that replaces the linear trend by a vector of year indicators. We also replace the policy indicator w_{it} by three dummy variables (for the years 2009, 2010, and 2011) that take the value of one if school i participates in the program. The coefficients of these dummy variables ($w_{1=2009}$, $w_{1=2010}$, and $w_{1=2011}$) give us information about the test score gains that result from each additional year of exposure to the program. We observe in table 6 that the rise in test scores is increasing. In math, there are no statistically significant gains in test scores after the first year. The second year of the program produces an additional gain of 4.1 points, whereas after the third year, test scores increase another 7.2 points. Overall, after 3 years, schools that participate in the program increase their math test scores by 11.2 points, which is equivalent to moving a school from the 50th percentile to the 65th percentile of the distribution. For language test scores, we observe no significant improvement after the first year, a rise of 4.0 points during the second year, and an additional gain of 3.9 points after the third year. The total gain in language test scores is 7.9 points, equivalent to moving a school from the 50th percentile to the 62nd percentile of the distribution. Those effects are equivalent to test score gains of 0.4 and 0.3 standard deviations in math and language, respectively.

Before presenting the results derived from the market-level analysis, we discuss some empirical issues regarding the difference-in-differences methodology and perform a robustness check of our main results. Tables are exhibited in the Appendix.

A first empirical concern that could emerge from the previous analysis is related to the validity of the control group. The analysis is based on the assumption that there are no unobservable time-varying factors correlated with the treatment status and also correlated with students' academic results. The inclusion of the fixed school effects in the empirical model removes any selection bias that arises from observed as well as unobserved time-invariant characteristics of schools but does not remove a bias that could arise from time-varying factors. Therefore, as a robustness analysis, we present an alternative methodology to build the treatment and control groups.

First, we show that the percentage of vulnerable students is positive and significantly correlated with the probability of enrolling in the program. We estimate a probit in which the dependent variable is the decision to join the program and the regressor is the percentage of vulnerable students enrolled in a school. Table A1 presents the results. We observe that

TABLE 6

Yearly Effects of the SEP Reform on School Academic Achievement

	SIMCE 4th-Grade Math				SIMCE 4th-Grade Language			
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$w_{1=2009}$	.958	1.4077**	.7281	.7388	1.1422*	1.4474**	.9546	.8912
	(.646)	(.633)	(.644)	(.634)	(.585)	(.576)	(.590)	(.588)
$w_{1=2010}$	4.0697***	4.8041***	3.9140***	3.9310***	3.9841***	4.4826***	3.8589***	3.7581***
	(.679)	(.672)	(.646)	(.652)	(.655)	(.660)	(.639)	(.648)
$w_{1=2011}$	11.2471***	11.9001***	10.7555***	10.7714***	7.9296***	8.3729***	7.5304***	7.4358***
	(.732)	(.725)	(.717)	(.712)	(.726)	(.723)	(.718)	(.717)
Gross family income		.0167***		.0004		.0114***		−.0022
		(.003)		(.003)		(.003)		(.003)
Mother's years of schooling			3.2126***	3.2075***			2.7196***	2.7499***
			(.288)	(.295)			(.257)	(.261)
Father's years of schooling			1.9060***	1.8993***			1.4689***	1.5087***
			(.252)	(.256)			(.255)	(.258)
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Source.—SIMCE 2006–11 and Ministry of Education, Chile.

Note.—$N = 18{,}203$. Gross family income corresponds to the average family income that students report at the moment of taking SIMCE. It is expressed in thousands of Chilean pesos (CLP) for each one of the years considered in the sample (1 US$ = 520 CLP). Standard errors (in parentheses) were estimated using clustering by municipality.

* $p < .1$.

** $p < .05$.

*** $p < .01$.

this pretreatment characteristic is positively correlated with the propensity to join the program. This result is indeed intuitive given the characteristics of the program. As discussed in Section III, schools that sign the agreement with the Ministry of Education and thus decide to participate in the program cannot charge copayments to priority students. That requirement automatically implies that participation in the program will be more costly for schools that high- or middle-income students attended during the pretreatment period (higher-copayment schools). Therefore, schools with an initially high percentage of vulnerable students should have had more incentives to join the program.[16]

Then we classify schools according to this pretreatment characteristic that is correlated with the propensity to join the program (instead of the actual decision itself). The cutoff to classify schools in the treatment and control groups is chosen to replicate the actual fraction of schools that join the program. Tables A2–A4 exhibit the results of regressions (1)–(3) under the new approach to build the control group. We still find a positive effect of targeted vouchers on standardized test scores under this alternative methodology to classify schools.[17]

A second empirical issue to be addressed is related to the change in the composition of the student body. It is important to examine whether the policy led to any noticeable compositional changes in the schools. The new sorting of children across schools would fundamentally complicate averaged school-level test scores, which reflect both school quality and student composition. The assumption of "parallel trends in covariates" implied by the difference-in-differences assumption would be violated.

The sign of the bias generated by the sorting of students depends on the type of sorting induced by the program, which depends heavily on some specific characteristics of the reform. Schools that receive funds from the program must exempt all priority students from paying any copayment and cannot select among those priority students. An important change in the observable composition of students in the treatment group and the control group would be produced if schools that initially enrolled middle- and high-income students (higher-copayment schools) joined the program and began to enroll priority students. We should point out that it is unlikely that those schools have large incentives to join the program since one of the main costs for schools derived from the Equality of Opportunity and Educational Excellence Agreement is the impossibility of charging private copayments to priority students. However, if this type of sorting existed, the socioeconomic composition of treatment schools should worsen and a difference-in-differences methodology would underestimate potential positive effects of the reform.

[16] Data from the Ministry of Education show that participating schools charged an average monthly copayment of $26 before the implementation of the reform, much lower than the $66 charged by nonparticipating schools.

[17] We are grateful to Eric Hanushek and an anonymous referee of this *Journal* for suggesting this alternative approach to classifying schools.

To examine whether the reform led to any noticeable compositional change in the control and treatment groups, we look at two pieces of evidence. First, we analyze the evolution over time of the average enrollment rate in treatment (control) schools. Second, we compare the trend followed by some socioeconomic indicators in the control and treatment groups. This information is useful because the main empirical concern should be about movements of students with some characteristics from a control (treatment) school to a treatment (control) school with a very different composition of students. To be clear, movements of students from school A to school B within the control group or within the treatment group do not introduce any bias in our estimations because the average result of the group (treatment or control) should not be affected. Similarly, movements of students with some characteristic to schools with a similar composition of students would not violate the assumption of parallel trends in covariates.

We observe in table A5 that the enrollment rate in the treatment and control groups remained pretty stable during the period 2006–11. Additionally, we do not observe any noticeable compositional changes in the control and treatment groups. For instance, during the pretreatment period, on average, treatment schools exhibited a gross family income 44 percent lower than that of treatment schools. During the posttreatment period, we observe only a slight worsening in the average gross family income of the treatment group relative to that of the control group (that evidence is consistent with the type of sorting discussed above). Therefore, we do not observe in the data any noticeable change in the composition of students in control and treatment schools. The absence of evident compositional changes is consistent with the fact that joining the reform is very costly for high- and middle-income schools because they cannot charge copayments to priority students.

As an additional robustness check, we estimate regressions (1)–(3) excluding schools that exhibited a significant change in the socioeconomic characteristics of their student body between the pre- and posttreatment periods. Specifically, we exclude schools in which the average gross family income of their students grew more (less) than one standard deviation above (below) the mean. Therefore, we estimate the empirical model including only schools in which compositional changes were small. Tables A6–A8 present the results, which confirm the positive effects of the program on standardized test scores.

Additionally, it could be argued that 2008 cannot be considered a pretreatment year since the targeted voucher program was created in 2008, and thus, schools already knew about the program. As a robustness analysis, we present in table A9 the results of regression (1) excluding the year 2008 from the pretreatment period (and keeping the posttreatment period as 2009–11). We observe that conclusions do not change.

It would also be relevant to discard the fact that the main result does not respond to an improvement in academic results before the implementation of the program. To check that it is not the case, we include in the empirical model dummy variables 1 and 2 years before the introduction of the reform that take the value of one for the treatment group. In this way, we test whether our results can be simply explained by program participant schools that changed their behavior before signing the Equality of Opportunity and Educational Excellence Agreement. We present the results in table A10. The policy indicators for the 2 years before the implementation of the program are not significant at conventional levels. Therefore, we can discard the fact that the main result responds to an improvement in academic results before the implementation of the program.

Finally, the last empirical concern to be addressed is related to the fact that school participation varies by school sector. Almost 90 percent of public schools joined the program, whereas only around 40 percent of subsidized private schools did. One potential reason behind this asymmetric participation by sectors is, again, related to one of the main characteristics of the reform. The program's mandatory prohibition on charging copayments to vulnerable students made participation in the program less costly for free public schools than for subsidized private schools that charge copayments. Given this difference in participation, we estimate regressions (1)–(3) considering only voucher private schools. Tables A11–A13 present the results. Again, we observe positive effects of targeted vouchers on standardized test scores.

Market-level analysis.—As discussed in Section V, we also perform a market-level analysis. To do so, we define each municipality as an individual market and compare the changes in test scores after the implementation of the reform in a market in which some schools signed the agreement and others did not. This empirical approach allows us to better capture competitive forces. Regarding competition for shares, the improvement in the academic results of the treatment group encourages the control group to react and compete and, thus, to potentially improve education quality. In that type of education market, a market-level analysis is a more compelling approach to estimate the effects of the reform on academic results than a difference-in-differences approach.

As we can see in table 7, from the coefficient of S_i, we can infer that 1 percent of schools joining the program produce a gain of 0.18 points in math and 0.11 points in language test scores in the average market. Since in our sample around 80 percent of the schools joined the program, we may observe average gains of 14.4 points in math and 8.8 points in language test scores after 3 years of exposure to the reform relative to a market in which the reform is not implemented. Those effects are equivalent to test score gains of 0.53 and 0.37 standard deviations in math and language, respectively. Therefore, this market-level analysis confirms the positive and statistically significant effects of targeted vouchers on school results.

Additionally, as a robustness analysis, we present a different definition of education markets. We group the municipalities of the three most densely

TABLE 7

Effects of the SEP Reform on School Academic Achievement: Market-Level Analysis

	SIMCE 4th-Grade Math					SIMCE 4th-Grade Language				
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
S_i	17.0557***	17.1119***	18.0110***	16.9264***	18.2149***	10.4087***	10.4490***	10.8473***	10.2728***	10.6930***
	(4.061)	(4.132)	(4.057)	(4.086)	(4.126)	(3.067)	(3.067)	(3.059)	(3.101)	(3.153)
Δ SIMCE (2006–8)		.0564	.0438	.0496	.0473		.0168	.0116	.0161	.0161
		(.072)	(.072)	(.075)	(.075)		(.059)	(.059)	(.061)	(.061)
Δ Gross family income (2006–8)			.0428*		.0417			.0193		.0136
			(.025)		(.030)			(.021)		(.024)
Δ Mother's years of schooling (2006–8)				3.058	1.8498				2.1857	1.7915
				(2.510)	(2.610)				(2.013)	(2.147)
Δ Father's years of schooling (2006–8)				−1.7721	−2.1429				−1.1152	−1.2378
				(2.323)	(2.353)				(1.687)	(1.739)
Constant	−6.6268**	−6.5631*	−8.2412**	−6.4938*	−8.4564**	−2.4551	−2.5805	−3.2916	−2.4771	−3.1143
	(3.275)	(3.347)	(3.361)	(3.314)	(3.538)	(2.438)	(2.451)	(2.503)	(2.501)	(2.715)

Source.—SIMCE 2006–11 and Ministry of Education, Chile.

Note.—$N = 309$. Gross family income corresponds to the average family income that students report at the moment of taking SIMCE. It is expressed in thousands of Chilean pesos (CLP) for each one of the years considered in the sample (1 US$ = 520 CLP). Standard errors (in parentheses) were estimated using clustering by municipality.

* $p < .1$.

** $p < .05$.

*** $p < .01$.

populated and urbanized provinces as individual markets (Valparaíso, Concepción, and Santiago). In those provinces, the mobility cost of students should be small; thus, we consider each of those provinces as individual markets. In more rural provinces, students cannot easily move to schools outside the municipality where their family lives. This constraint should strongly decrease the demand for more distant schools, as suggested by Gallego and Hernando (2009). Table A14 in the Appendix exhibits the results. We observe that our conclusions are robust to this alternative definition of education markets.[18]

VII. Discussion

The SEP reform is worthy of evaluation because of the conjunction of four factors: (1) the program delivers more resources to the education market, (2) those resources are conditional on the accomplishment of some specific scholastic goals, (3) those resources are allocated in a market in which public and private providers compete, and (4) the resources are allocated in a market in which some institutional elements prevent public schools from behaving completely as profit-maximizing firms. On the basis of the empirical results presented in this paper, along with some institutional characteristics that might appear in a mixed-funding school system, we can speculate about the more likely characterization of the school market and the channels through which the type of reform analyzed in this paper may affect the incentives of schools. However, a formal empirical analysis to disentangle the specific channels through which conditional vouchers encourage schools to offer a higher quality of education constitutes an important avenue for future research.

The Chilean school system is a mixed-funding system in which we can find four types of schools: (i) "residual" public schools, where the children who attend are those who either cannot afford fee-charging voucher private schools or are denied admission to private schools;[19] (ii) free private voucher schools, which offer a lower quality and compete for low-income students who cannot afford private copayments; (iii) fee-charging voucher private schools, which supply a higher education quality; and (iv) elite private schools, which do not receive public funds.

A voucher system should encourage schools that enroll low-income students to compete for these resources. In an environment with some mobility of students across schools, voucher private schools behave very

[18] Estimates using each province as an individual market are unfeasible because we are left with a small number of observations. Additionally, that definition of individual markets would not be completely convincing for reasons previously explained.

[19] In this regard, the SEP reform is different from the universal voucher program introduced in 1981. Whereas private schools can employ competitive admission processes to admit only the most promising applicants among those who receive funds from the universal voucher program, they cannot select among priority students who receive SEP funds. Public schools with vacancies are legally obliged to admit all applicants regardless of their academic potential, socioeconomic backgrounds, or the type of voucher or financing that they have.

much like profit-maximizing firms. Competition for the voucher resources may give incentives to voucher private schools to attract low-income students and, thus, to offer a higher level of quality.[20] In principle, public schools should react by also increasing their quality. However, some institutional characteristics of public schools prevent them from behaving in a fully profit-maximizing manner.

The Teaching Statute constrains public schools from freely competing in the education market. Public schools cannot easily dismiss teachers or modify the structure of monetary incentives for their teachers. They have more flexibility to implement reforms in other dimensions, such as infrastructure, curriculum management, or student life. However, there is a second element that could slow the progress of public schools even in those areas. Several public schools receive nonvoucher transfers from municipalities. These are the so-called soft budget constraints (Sapelli and Vial 2002; Hsieh and Urquiola 2006). These nonvoucher transfers introduce perverse incentives since schools that experience a decrease in their enrollment do not experience a worsening in their budgets. Therefore, public schools that face strong competition from voucher private schools do not react by increasing quality, even though they lose students. Nonvoucher transfers from municipalities allow them to continue operating and prevent a closure of these schools.[21] Figure 2 shows the average annual nonvoucher transfers per student in different municipalities.[22] We observe that nonvoucher transfers are present in practically all municipalities, although a lot of heterogeneity is observed.

All these elements decrease the degree to which unconditional vouchers encourage public schools in Chile to behave as profit-maximizing firms. Therefore, it is not straightforward that unconditional vouchers should have led public schools to improve their quality. However, since the SEP program does consider conditionality in the allocation of vouchers, this conditionality might be important to understand why the SEP reform produced academic improvements in public schools. The Equality of Opportunity and Educational Excellence Agreement requires schools to implement education reforms, and noncompliance with those reforms can result in the cancellation of the official recognition given by the Ministry of Education. The required reforms by the Equality of Opportunity and Educational Excellence Agreement force schools to progress in some specific areas, undermining in this way the negative incentives

[20] If competition existed only among treatment schools (e.g., as a consequence of a very segregated education market), our difference-in-differences methodology would be compelling to capture the impact of the reform. If, in addition, control schools reacted to the rise in quality in treatment schools, the market-level analysis would deliver a more robust estimation of the effects of the reform.

[21] For instance, Hsieh and Urquiola (2006) show that despite extensive private entry and sustained declines in public enrollments, the aggregate number of municipal schools has barely fallen.

[22] Nine municipalities were excluded from the graph for expositional reasons. Those municipalities present an average nonvoucher transfer between $2,000 and $15,000.

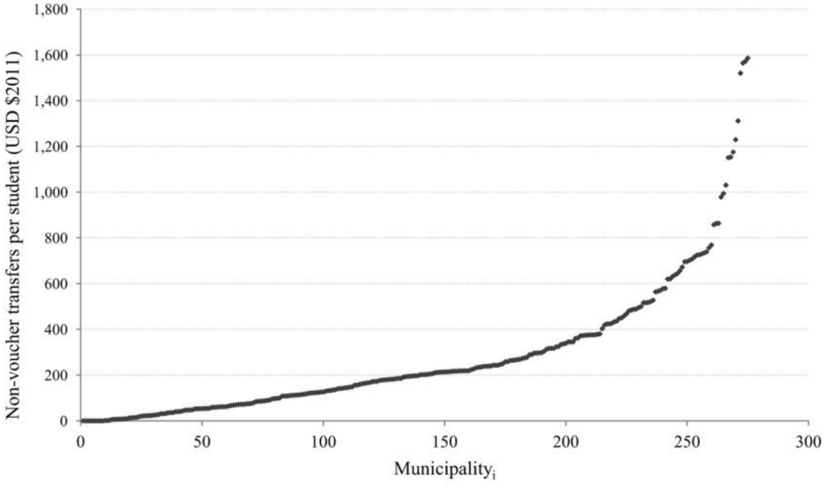

Figure 2.—Nonvoucher transfers at the municipal level

derived from the nonvoucher transfers. Even though public schools are constrained by the Teaching Statute, there are several other areas in which schools can experience improvements. Additionally, the SEP law establishes some flexibility to hire teachers, education assistants, and the necessary staff to improve the technical and pedagogical performance of the schools.

Our results are consistent with a story in which the SEP reform delivered extra resources to the market, targeted to low-income students. Competition for those resources encouraged private voucher schools to offer a higher quality to attract more priority students. Public schools should have reacted by increasing their quality as long as the enrollment elasticity was not so low. However, the Teaching Statute and the presence of soft budget constraints in public schools prevented, to some degree, profit-maximizing behavior. At the same time, these specific institutional characteristics of public schools introduced an important role for the conditionality embodied in the Equality of Opportunity and Educational Excellence Agreement. That agreement forces schools to progress in reforms aimed at improving the academic achievement of their students. Therefore, public schools that signed the agreement experienced an improvement in their quality, although perhaps smaller than that of the private voucher schools.

VIII. Conclusions

This paper provides empirical evidence of the effect of targeted vouchers on school results. We use data from the targeted voucher program implemented in Chile in 2008. This program is different in several dimensions from the universal voucher program implemented in 1981.

First, this voucher program is focused on low-income students. Second, the program requires that schools sign the Equality of Opportunity and Educational Excellence Agreement to receive the extra resources delivered by the program. That agreement forces schools to present an Educational Improvement Plan to the Ministry of Education, which details specific education reforms that the school must undertake to improve results on the SIMCE, a national standardized test. Third, schools must also exempt all priority students from paying any copayment or tuition. Finally, schools joining the program cannot select among priority students.

Using a difference-in-differences approach and a market-level analysis, we find positive and statistically significant effects of vouchers on math and language test scores. After 3 years of exposure to the reform, the difference-in-differences approach shows that participating schools increased their math test scores by 0.4 standard deviations, which is equivalent to moving a school from the 50th percentile to the 65th percentile of the distribution. The total gain in language test scores was 0.3 standard deviations, equivalent to moving a school from the 50th percentile to the 62nd percentile of the distribution. The market-level analysis shows that for each 1 percent of schools joining the program, the average market experiences test score gains of 0.18 points in math and 0.11 points in language. This means that, after 3 years, a market in which 80 percent of schools participate in the reform experiences average gains of 14.4 points in math and 8.8 points in language, relative to a market in which the reform is not implemented. Those effects are equivalent to test score gains of 0.53 and 0.37 standard deviations in math and language, respectively. The positive effect of the program on standardized test scores is robust to different alternative empirical specifications.

Our results highlight the importance of conditioning the delivery of resources to some specific academic goals in markets with institutional characteristics that prevent public schools from behaving as profit-maximizing firms. Further analysis to disentangle the mechanisms through which targeted vouchers change the education market constitutes an interesting and important avenue for future research.

Appendix

Check of Robustness

TABLE A1
PROBABILITY OF ENROLLING IN THE PROGRAM

	(1)	(2)	(3)	(4)
% of vulnerable students 2006	.1042***			
	(.004)			
% of vulnerable students 2007		.1201***		
		(.005)		
% of vulnerable students 2008			.0462***	
			(.002)	
% of vulnerable students 2006–8 (average)				.0852***
				(.003)
Constant	−.4136***	−.3528***	−1.0282***	−.9145***
	(.045)	(.044)	(.061)	(.058)
Pseudo R^2	.37	.37	.42	.45

Source.—SIMCE 2006–11 and Ministry of Education, Chile.
Note.—$N = 3{,}033$. Standard errors are in parentheses.
* $p < .1$.
** $p < .05$.
*** $p < .01$.

TABLE A2

Effects of the SEP Reform on School Academic Achievement, Alternative Definition of Treatment and Control Groups

	SIMCE 4th-Grade Math				SIMCE 4th-Grade Language			
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
λ_t	.8704***	.7390***	.8382***	.9232***	2.3565***	2.1863***	2.3276***	2.3213***
	(.137)	(.148)	(.132)	(.149)	(.106)	(.105)	(.101)	(.104)
w_{it}	4.7220***	4.9069***	4.4127***	4.2797***	3.2734***	3.5129***	3.0304***	3.0404***
	(.433)	(.432)	(.415)	(.423)	(.393)	(.381)	(.384)	(.383)
Gross family income		.0103***		−.0067**		.0133***		.0005
		(.003)		(.003)		(.003)		(.003)
Mother's years of schooling			3.2540***	3.3438***			2.7384***	2.7317***
			(.289)	(.295)			(.259)	(.263)
Father's years of schooling			2.0069***	2.1446***			1.4551***	1.4462***
			(.256)	(.261)			(.249)	(.254)
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Source.—SIMCE 2006–11 and Ministry of Education, Chile.

Note.—$N = 18{,}203$. We use an alternative definition of treatment and control groups. We classify schools according to the percentage of vulnerable students, which is positive and significantly correlated with the probability of enrolling in the program. The cutoff was chosen to replicate the actual fraction of schools that join the program. Gross family income corresponds to the average family income that students report at the moment of taking SIMCE. It is expressed in thousands of Chilean pesos (CLP) for each one of the years considered in the sample (1 US$ = 520 CLP). Standard errors (in parentheses) were estimated using clustering by municipality.

* $p < .1$.

** $p < .05$.

*** $p < .01$.

TABLE A3

Cumulative Effects of the SEP Reform on School Academic Achievement, Alternative Definition of Treatment and Control Groups

	SIMCE 4th-Grade Math				SIMCE 4th-Grade Language			
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
λ_t	−.1452	−.4069***	−.1037	−.1179	2.1269***	1.8872***	2.1581***	2.1166***
	(.133)	(.137)	(.127)	(.140)	(.108)	(.105)	(.104)	(.106)
$\lambda_t' \times w_{it}$	4.0221***	4.2394***	3.7534***	3.7666***	1.8117***	2.0107***	1.5918***	1.6300***
	(.231)	(.224)	(.218)	(.217)	(.191)	(.185)	(.188)	(.187)
Gross family income		.0170***		.0009		.0156***		.0026
		(.003)		(.003)		(.003)		(.003)
Mother's years of schooling			3.1028***	3.0901***			2.6855***	2.6487***
			(.290)	(.297)			(.259)	(.263)
Father's years of schooling			1.8963***	1.8800***			1.4196***	1.3721***
			(.254)	(.259)			(.248)	(.253)
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Source.—SIMCE 2006–11 and Ministry of Education, Chile.

Note.—$N = 18{,}203$. We use an alternative definition of treatment and control groups. We classify schools according to the percentage of vulnerable students, which is positive and significantly correlated with the probability of enrolling in the program. The cutoff was chosen to replicate the actual fraction of schools that join the program. Gross family income corresponds to the average family income that students report at the moment of taking SIMCE. It is expressed in thousands of Chilean pesos (CLP) for each one of the years considered in the sample (1 US$ = 520 CLP). Standard errors (in parentheses) were estimated using clustering by municipality.

* $p < .1$.

** $p < .05$.

*** $p < .01$.

TABLE A4

Yearly Effects of the SEP Reform on School Academic Achievement, Alternative Definition of Treatment and Control Groups

	SIMCE 4th-Grade Math				SIMCE 4th-Grade Language			
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$w_{1=2009}$	1.2204*	1.7097**	1.0708	1.0881	1.7099***	2.0768***	1.5870**	1.5642**
	(.700)	(.694)	(.689)	(.692)	(.624)	(.621)	(.614)	(.621)
$w_{1=2010}$	3.8658***	4.6946***	3.7022***	3.7311***	5.3302***	5.9516***	5.1948***	5.1566***
	(.708)	(.701)	(.668)	(.681)	(.605)	(.598)	(.582)	(.590)
$w_{1=2011}$	11.0650***	11.7584***	10.4276***	10.4537***	8.5099***	9.0298***	7.9910***	7.9564***
	(.775)	(.774)	(.742)	(.761)	(.730)	(.729)	(.706)	(.718)
Gross family income		.0169***		.0006		.0127***		−.0008
		(.003)		(.003)		(.003)		(.003)
Mother's years of schooling			3.1713***	3.1634***			2.6822***	2.6927***
			(.288)	(.295)			(.258)	(.262)
Father's years of schooling			1.9027***	1.8924***			1.4694***	1.4829***
			(.251)	(.257)			(.254)	(.258)
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Source.—SIMCE 2006–11 and Ministry of Education, Chile.

Note.—$N = 18{,}203$. We use an alternative definition of treatment and control groups. We classify schools according to the percentage of vulnerable students, which is positive and significantly correlated with the probability of enrolling in the program. The cutoff was chosen to replicate the actual fraction of schools that join the program. Gross family income corresponds to the average family income that students report at the moment of taking SIMCE. It is expressed in thousands of Chilean pesos (CLP) for each one of the years considered in the sample (1 US$ = 520 CLP). Standard errors (in parentheses) were estimated using clustering by municipality.

* $p < .1$.

** $p < .05$.

*** $p < .01$.

TABLE A5
Evolution of the Enrollment Rate and Socioeconomic Variables in the Treatment and Control Groups

	2006	2007	2008	2009	2010	2011
Enrollment rate (treatment)	74.6	73.7	72.9	72.2	71.7	71.20
Enrollment rate (control)	25.4	26.3	27.1	27.8	28.3	28.8
Ratio of gross family income (treatment/control)	.46	.46	.46	.45	.45	.45
Difference in mother's years of schooling (treatment − control)	−2.9	−2.9	−2.9	−2.9	−2.9	−2.8
Difference in father's years of schooling (treatment −control)	−3.0	−3.0	−3.0	−3.0	−3.0	−2.9

Source.—SIMCE 2006–11 and Ministry of Education, Chile.
Note.—Gross family income corresponds to the average family income that students report at the moment of taking SIMCE. It is expressed in thousands of Chilean pesos (CLP) for each one of the years considered in the sample (1 US$ = 520 CLP).

TABLE A6

Effects of the SEP Reform on School Academic Achievement, Restricted Sample

Variable	SIMCE 4th-Grade Math (1)	(2)	(3)	(4)	SIMCE 4th-Grade Language (5)	(6)	(7)	(8)
λ_t	.7651***	.6667***	.7693***	.8422***	2.3538***	2.2026***	2.3526***	2.3307***
	(.136)	(.147)	(.132)	(.147)	(.110)	(.110)	(.107)	(.108)
w_{it}	4.7942***	4.9497***	4.622***	4.5004***	2.9166***	3.1552***	2.7828***	2.8193***
	(.470)	(.467)	(.461)	(.458)	(.457)	(.450)	(.451)	(.445)
Gross family income		.0100***		−.0073**		.0153***		.0022
		(.003)		(.004)		(.003)		(.003)
Mother's years of schooling			3.425***	3.5099***			2.9421***	2.9166***
			(.311)	(.315)			(.283)	(.287)
Father's years of schooling			1.9350***	2.0521***			1.4007***	1.3654***
			(.276)	(.280)			(.275)	(.279)
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Source.—SIMCE 2006–11 and Ministry of Education, Chile.

Note.—$N = 15{,}630$. The restricted sample excludes schools that exhibited a significant change in the socioeconomic characteristics of their student body between the pre- and posttreatment periods. Specifically, we exclude schools in which the average gross family income of their students grew more (less) than one standard deviation above (below) the mean. Gross family income corresponds to the average family income that students report at the moment of taking SIMCE. It is expressed in thousands of Chilean pesos (CLP) for each one of the years considered in the sample (1 US$ = 520 CLP). Standard errors (in parentheses) were estimated using clustering by municipality.

* $p < .1$.

** $p < .05$.

*** $p < .01$.

TABLE A7

CUMULATIVE EFFECTS OF THE SEP REFORM ON SCHOOL ACADEMIC ACHIEVEMENT, RESTRICTED SAMPLE

	SIMCE 4th-Grade Math				SIMCE 4th-Grade Language			
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
λ_t	−.2775**	−.4675***	−.1901	−.1885	2.1430***	1.9423***	2.2131***	2.1653***
	(.127)	(.134)	(.125)	(.138)	(.116)	(.116)	(.114)	(.115)
$\lambda_t' \times w_{it}$	4.1046***	4.2710***	3.8646***	3.8631***	1.6265***	1.8022***	1.4264***	1.4715***
	(.236)	(.230)	(.232)	(.228)	(.216)	(.213)	(.218)	(.215)
Gross family income		.0161***		−.0001		.0170***		.0038
		(.003)		(.004)		(.003)		(.003)
Mother's years of schooling			3.2692***	3.2707***			2.8821***	2.8356***
			(.311)	(.317)			(.282)	(.286)
Father's years of schooling			1.7781***	1.7801***			1.3631***	1.2996***
			(.274)	(.277)			(.276)	(.278)
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Source.—SIMCE 2006–11 and Ministry of Education, Chile.

Note.—$N = 15{,}630$. The restricted sample excludes schools that exhibited a significant change in the socioeconomic characteristics of their student body between the pre- and posttreatment periods. Specifically, we exclude schools in which the average gross family income of their students grew more (less) than one standard deviation above (below) the mean. Gross family income corresponds to the average family income that students report at the moment of taking SIMCE. It is expressed in thousands of Chilean pesos (CLP) for each one of the years considered in the sample (1 US$ = 520 CLP). Standard errors (in parentheses) were estimated using clustering by municipality.

* $p < .1$.

** $p < .05$.

*** $p < .01$.

TABLE A8

Yearly Effects of the SEP Reform on School Academic Achievement, Restricted Sample

	SIMCE 4th-Grade Math				SIMCE 4th-Grade Language			
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$w_{1=2009}$	.9195	1.2857*	.7051	.7221	1.2313*	1.4955**	1.0527	1.0182
	(.739)	(.718)	(.735)	(.715)	(.694)	(.679)	(.697)	(.685)
$w_{1=2010}$	4.3350***	4.9498***	4.1824***	4.2102**	4.0494***	4.4932***	3.9246***	3.8683***
	(.775)	(.759)	(.737)	(.740)	(.710)	(.707)	(.689)	(.694)
$w_{1=2011}$	11.4417***	11.9529***	10.9623***	10.9868***	8.1605***	8.5294***	7.7620***	7.7121***
	(.773)	(.765)	(.763)	(.757)	(.776)	(.767)	(.765)	(.760)
Gross family income		.0172***		.0008		.0124***		−.0015
		(.003)		(.004)		(.003)		(.003)
Mother's years of schooling			3.3445***	3.3354***			2.9028***	2.9211***
			(.308)	(.314)			(.281)	(.285)
Father's years of schooling			1.8034***	1.7910***			1.4170***	1.4422***
			(.273)	(.276)			(.284)	(.285)
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Source.—SIMCE 2006–11 and Ministry of Education, Chile.

Note.—N = 15,630. The restricted sample excludes schools that exhibited a significant change in the socioeconomic characteristics of their student body between the pre- and posttreatment periods. Specifically, we exclude schools in which the average gross family income of their students grew more (less) than one standard deviation above (below) the mean. Gross family income corresponds to the average family income that students report at the moment of taking SIMCE. It is expressed in thousands of Chilean pesos (CLP) for each one of the years considered in the sample (1 US$ = 520 CLP). Standard errors (in parentheses) were estimated using clustering by municipality.

* $p < .1$.

** $p < .05$.

*** $p < .01$.

TABLE A9

EFFECTS OF THE SEP REFORM ON SCHOOL ACADEMIC ACHIEVEMENT, EXCLUDING YEAR 2008

	SIMCE 4th-Grade Math				SIMCE 4th-Grade Language			
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
λ_t	1.1366***	1.0606***	1.0458***	1.1814***	2.1825***	2.0111***	2.1038***	2.1016***
	(.157)	(.173)	(.150)	(.168)	(.116)	(.120)	(.112)	(.118)
w_{it}	3.0397***	3.1647***	3.0983***	2.8701***	4.5049***	4.7869***	4.5569***	4.5607***
	(.554)	(.564)	(.537)	(.546)	(.493)	(.492)	(.480)	(.481)
Gross family income		.0057*		−.0105***		.0129***		.0002
		(.003)		(.003)		(.003)		(.003)
Mother's years of schooling			3.2780***	3.4125***			2.8531***	2.8509***
			(.335)	(.340)			(.290)	(.293)
Father's years of schooling			1.8276***	2.0168***			1.4422***	1.4391***
			(.287)	(.291)			(.281)	(.284)
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Source.—SIMCE 2006–11 and Ministry of Education, Chile.

Note.—$N = 15{,}169$. Gross family income corresponds to the average family income that students report at the moment of taking SIMCE. It is expressed in thousands of Chilean pesos (CLP) for each one of the years considered in the sample (1 US$ = 520 CLP). Standard errors (in parentheses) were estimated using clustering by municipality.

* $p < .1$.

** $p < .05$.

*** $p < .01$.

TABLE A10
EFFECTS OF THE SEP REFORM ON SCHOOL ACADEMIC ACHIEVEMENT, ADDITIONAL POLICY INDICATORS

	SIMCE 4th-Grade Math				SIMCE 4th-Grade Language			
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$w_{1=2007}$	−.9758	−.9119	−1.1696	−1.1679	−.5203	−.4762	−.6789	−.6892
	(.783)	(.793)	(.780)	(.784)	(.618)	(.625)	(.616)	(.617)
$w_{1=2008}$	−.4773	−.2196	−.5290	−.5238	.9119	1.0897	.8719	.8391
	(.839)	(.840)	(.833)	(.836)	(.713)	(.712)	(.709)	(.706)
$w_{1=2009}$	.4737	1.0308	.1617	.1733	1.2728*	1.6572**	1.0188	.9460
	(.772)	(.765)	(.775)	(.778)	(.712)	(.702)	(.721)	(.722)
$w_{1=2010}$	3.5854***	4.4272***	3.3477***	3.3648***	4.1147***	4.6957***	3.9232***	3.8159***
	(.790)	(.800)	(.754)	(.778)	(.788)	(.798)	(.772)	(.784)
$w_{1=2011}$	10.7627***	11.5232***	10.1889***	10.2051***	8.0602***	8.5851***	7.5945***	7.4929***
	(.849)	(.858)	(.845)	(.856)	(.831)	(.831)	(.836)	(.838)
Gross family income		.0167***		.0003		.0116***		−.0021
		(.003)		(.003)		(.003)		(.003)
Mother's years of schooling			3.2148***	3.2104***			2.7237***	2.7516***
			(.288)	(.295)			(.258)	(.262)
Father's years of schooling			1.9076***	1.9017***			1.4680***	1.5047***
			(.252)	(.256)			(.255)	(.257)
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Source.—SIMCE 2006–11 and Ministry of Education, Chile.
Note.—$N = 18{,}203$. Gross family income corresponds to the average family income that students report at the moment of taking SIMCE. It is expressed in thousands of Chilean pesos (CLP) for each one of the years considered in the sample (1 US$ = 520 CLP). Standard errors (in parentheses) were estimated using clustering by municipality.
* $p < .1$.
** $p < .05$.
*** $p < .01$.

TABLE A11

EFFECTS OF THE SEP REFORM ON SCHOOL ACADEMIC ACHIEVEMENT, EXCLUDING PUBLIC SCHOOLS

	SIMCE 4th-Grade Math				SIMCE 4th-Grade Language			
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
λ_t	.5162***	.3657**	.4945***	.5331***	2.0932***	1.9483***	2.0717***	2.0685***
	(.127)	(.148)	(.124)	(.152)	(.115)	(.117)	(.112)	(.118)
w_{it}	5.5154***	5.7414***	5.4242***	5.3634***	3.6935***	3.9112***	3.6261***	3.6312***
	(.534)	(.523)	(.521)	(.510)	(.540)	(.530)	(.536)	(.532)
Gross family income		.0099***		−.0025		.0095***		.0002
		(.003)		(.004)		(.003)		(.003)
Mother's years of schooling			2.9579***	3.0087***			2.5048***	2.5006***
			(.373)	(.395)			(.348)	(.357)
Father's years of schooling			1.6374***	1.7064***			1.0854***	1.0797***
			(.318)	(.328)			(.340)	(.348)
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Source.—SIMCE 2006–11 and Ministry of Education, Chile.

Note.—$N = 9{,}312$. Gross family income corresponds to the average family income that students report at the moment of taking SIMCE. It is expressed in thousands of Chilean pesos (CLP) for each one of the years considered in the sample (1 US$ = 520 CLP). Standard errors (in parentheses) were estimated using clustering by municipality.

* $p < .1$.

** $p < .05$.

*** $p < .01$.

TABLE A12

CUMULATIVE EFFECTS OF THE SEP REFORM ON SCHOOL ACADEMIC ACHIEVEMENT, EXCLUDING PUBLIC SCHOOLS

	SIMCE 4th-Grade Math				SIMCE 4th-Grade Language			
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
λ_t	.2140*	.0101	.2223*	.2155	2.0714***	1.9036***	2.0737***	2.0595***
	(.124)	(.141)	(.122)	(.146)	(.123)	(.125)	(.120)	(.126)
$\lambda_t' \times w_{it}$	3.1261***	3.2885***	3.0056***	3.0114***	1.5739***	1.7076***	1.4794***	1.4916***
	(.272)	(.262)	(.266)	(.255)	(.262)	(.259)	(.262)	(.261)
Gross family income		.0123***		.0004		.0101***		.0008
		(.003)		(.004)		(.003)		(.003)
Mother's years of schooling			2.8610***	2.8528***			2.4563***	2.4391***
			(.375)	(.397)			(.351)	(.361)
Father's years of schooling			1.5770***	1.5660***			1.0677***	1.0445***
			(.316)	(.324)			(.341)	(.349)
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Source.—SIMCE 2006–11 and Ministry of Education, Chile.

Note.—$N = 9{,}312$. Gross family income corresponds to the average family income that students report at the moment of taking SIMCE. It is expressed in thousands of Chilean pesos (CLP) for each one of the years considered in the sample (1 US$ = 520 CLP). Standard errors (in parentheses) were estimated using clustering by municipality.

* $p < .1$.

** $p < .05$.

*** $p < .01$.

TABLE A13

Yearly Effects of the SEP Reform on School Academic Achievement, Excluding Public Schools

Variable	SIMCE 4th-Grade Math				SIMCE 4th-Grade Language			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$w_{1=2009}$	1.7785**	2.1096***	1.7818**	1.7895**	1.5655**	1.7308**	1.5682**	1.4711**
	(.782)	(.761)	(.778)	(.758)	(.734)	(.722)	(.736)	(.734)
$w_{1=2010}$	4.0890***	4.6016***	4.1034***	4.1153***	4.1573***	4.4132***	4.1718***	4.0211***
	(.841)	(.834)	(.815)	(.808)	(.749)	(.755)	(.727)	(.736)
$w_{1=2011}$	9.6346***	10.0940***	9.4234***	9.4347***	7.0138***	7.2431***	6.8526***	6.7091***
	(.910)	(.882)	(.889)	(.863)	(.897)	(.885)	(.893)	(.888)
Gross family income		.0125***		.0003		.0062**		−.0037
		(.003)		(.004)		(.003)		(.003)
Mother's years of schooling			2.9443***	2.9385***			2.4489***	2.5228***
			(.371)	(.391)			(.349)	(.360)
Father's years of schooling			1.5997***	1.5918***			1.0786***	1.1783***
			(.316)	(.322)			(.340)	(.349)
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Source.—SIMCE 2006–11 and Ministry of Education, Chile.

Note.—$N = 9{,}312$. Gross family income corresponds to the average family income that students report at the moment of taking SIMCE. It is expressed in thousands of Chilean pesos (CLP) for each one of the years considered in the sample (1 US$ = 520 CLP). Standard errors (in parentheses) were estimated using clustering by municipality.

* $p < .1$.

** $p < .05$.

*** $p < .01$.

TABLE A14

Effects of the SEP Reform on School Academic Achievement: Market-Level Analysis, Alternative Definition of Education Markets

	SIMCE 4th-Grade Math					SIMCE 4th-Grade Language				
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
S_i	15.0376***	15.0437**	16.1524***	14.9694***	16.5990***	11.3027***	11.3390***	12.0071***	11.3186***	12.3348***
	(4.894)	(5.036)	(4.855)	(4.950)	(4.954)	(3.634)	(3.652)	(3.624)	(3.710)	(3.797)
Δ SIMCE (2006–8)		.0734	.0560	.0673	.0622		.0256	.0130	.0299	.0266
		(.074)	(.074)	(.077)	(.078)		(.062)	(.063)	(.065)	(.066)
Δ gross family income (2006–8)			.0549**		.0538			.0340		.0338
			(.027)		(.033)			(.024)		(.028)
Δ mother's years of schooling (2006–8)				3.6974	2.0347				2.1523	1.1007
				(2.563)	(2.742)				(2.182)	(2.376)
Δ father's years of schooling (2006–8)				−2.4223	−2.7334				−1.4557	−1.6400
				(2.550)	(2.541)				(1.873)	(1.886)
Constant	−4.7472	−4.5935	−6.6756	−4.7075	−7.1548	−3.2786	−3.4401	−4.6245	−3.5413	−5.0475
	(4.162)	(4.307)	(4.218)	(4.243)	(4.413)	(3.037)	(3.063)	(3.099)	(3.144)	(3.366)

Source.—SIMCE 2006–11 and Ministry of Education, Chile.

Note.—$N = 254$. We present a different definition of education markets. We group the municipalities of the three most densely populated and urbanized provinces as individual markets (Valparaíso, Concepción, and Santiago). Gross family income corresponds to the average family income that students report at the moment of taking SIMCE. It is expressed in thousands of Chilean pesos (CLP) for each one of the years considered in the sample (1 US$ = 520 CLP). Standard errors (in parentheses) were estimated using clustering by municipality.

* $p < .1$.

** $p < .05$.

*** $p < .01$.

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